

# ANUBHAV AGRAWAL

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## SUMMARY

ML Engineer with **3+ years** owning AI products end-to-end — from model design to real-time production deployment and revenue. Shipped a **\$200K+ ARR** computer-vision product line (**600+ FPS**, **3 patents**), then built the **retrieval and evaluation infrastructure** for a production LLM system — cutting query latency **68%** (**2.5s**  $\rightarrow$  **0.8s**) for **500+ concurrent users**.

## PROFESSIONAL EXPERIENCE

### Griphic

Delhi

#### AI Engineer - 2

Jan 2026 – Present

- Engineered the **adaptive-intelligence core**: a stateful **LangGraph** agent that scores responses, scales question **difficulty** live, and retrieval-conditions each *counter* and *follow-up probe* into an interviewer-grade assessment.
- Architected the **RAG pipeline** with **3 parallel retrieval paths** (conversation embeddings, MongoDB, knowledge vault), *hybrid keyword + dense search* and *cross-encoder re-ranking*; reached **0.95 Recall@5** over **10K+ docs** using **512-d** embeddings.
- Cut query latency **2.5s**  $\rightarrow$  **0.8s** (**68%**) via context-fusion & async orchestration; raised domain-query **F1** by **42%** on a held-out eval set *vs.* single-path baseline; serves **500+ users** at **99.9% uptime**.
- Stood up an **eval harness** (recall, faithfulness, answer-relevance) and **LLM guardrails** with citations for safe, grounded outputs.

### Innvolution Healthcare Pvt. Ltd.

Bangalore

#### Machine Learning Engineer & Team Lead

Sep 2023 – Dec 2025

- Owned **AIMAG** — data, model R&D, *quantization*, and release — an X-ray super-resolution & denoising product at **\$200K+ ARR**.
- Built the serving stack: **PyTorch**  $\rightarrow$  **ONNX**  $\rightarrow$  **TensorRT** with **INT8/FP16 quantization** + **FastAPI (C++)**; **600+ FPS** on RTX 4090 (**70% latency cut**), multi-scale upsampling at **<5% loss** and **35% lower VRAM**.
- Built an automatic **QCA** pipeline (multi-stage DL: *segmentation*  $\rightarrow$  *calibration*  $\rightarrow$  *prediction*) delivering measurements in **<4 s**.
- Fine-tuned **U-Net++** (**ResNet50** + attention) for coronary segmentation at **0.94 Dice**.
- Led a **6-engineer ML team**, owning roadmap, review standards, and the lifecycle through **clinical validation**.

#### Image Quality Engineer Intern

Nov 2022 – May 2023

- Built motion-compensated stitching & denoising for **DSA** sequences — **20% clarity gain** at **<2-pixel** alignment error via *optical-flow* registration.

## EDUCATION

### Birla Institute of Technology And Science (BITS) - Pilani

M.Tech in Artificial Intelligence & Machine Learning

2025 – 2027

### Bhilai Institute of Technology, Durg

B.Tech in Computer Science Engineering

2019 – 2023

## TECHNICAL SKILLS

**Languages:** Python, C++

**ML & Deep Learning:** PyTorch, TensorFlow, OpenCV, Scikit-Learn, XGBoost, NumPy, Pandas, Albumentations, TorchVision, SciPy

**LLMs & RAG:** LangGraph, LangChain, HuggingFace Transformers, Vector Search, Cross-Encoder Re-ranking, Embeddings

**Inference & Serving:** TensorRT, ONNX, INT8/FP16 Quantization, CUDA, FastAPI, Docker

**Tools:** Git, Linux

## PATENTS APPLIED

**AI-POWERED X-RAY IMAGE ENHANCEMENT** (202541071194) — Single-image blind super-resolution & denoising.

**ENHANCING BODY PART FROM ANGIOGRAPHY LOOP** (202441009957) — Dynamic roadmap generation from DSA sequences.

**IMAGE ENHANCEMENT IN X-RAY FLUOROSCOPY** (202541041511) — Real-time contrast enhancement & denoising.